

Name: Maitele Funzani

Student No: ST10547359

<https://github.com/maitelle-cell/PDAN8411>

ICE TASK 3: PIPELINES AND TEXT DATA

Amazon Fine Food Reviews:

1.

Online customer reviews give information, about how customers feel about products.. The huge number of reviews that come from shopping sites makes it hard for businesses to read each comment by hand. Text analytics and machine learning help to look at reviews at once and find patterns in how customers talk.

This project looks at the Amazon Fine Food Reviews data to answer two questions. First what do customer reviews talk about? Second what is the overall sentiment that customers give toward the products?

The analysis uses Python tools for cleaning text looking at the data creating TF-IDF features and running a Logistic Regression model to classify sentiment. Topic modelling with Latent Dirichlet Allocation (LDA) is also planned to find themes in customer reviews.

The main goal is to turn customer review text into information that helps watch product quality improve customer service and guide business decisions.

2. Dataset Overview

2.1 Dataset Description

[35]:

	Id	HelpfulnessNumerator	HelpfulnessDenominator	Score	Time
count	568454.000000	568454.000000	568454.000000	568454.000000	5.684540e+05
mean	284227.500000	1.743817	2.22881	4.183199	1.296257e+09
std	164098.679298	7.636513	8.28974	1.310436	4.804331e+07
min	1.000000	0.000000	0.00000	1.000000	9.393408e+08
25%	142114.250000	0.000000	0.00000	4.000000	1.271290e+09
50%	284227.500000	0.000000	1.00000	5.000000	1.311120e+09
75%	426340.750000	2.000000	2.00000	5.000000	1.332720e+09
max	568454.000000	866.000000	923.00000	5.000000	1.351210e+09

The analysis uses the Amazon Fine Food Reviews dataset. This set holds customer reviews of food items bought on Amazon. Key fields are the review text, product details, customer rating and other review data.

The data in this notebook has **568,454 reviews**. That many reviews give customer feedback for text analysis and machine learning.

Because the data is big and varied it is good for spotting repeated customer concerns and good experiences.

2.2 Dataset Relevance

The data is directly useful for this business case because it shows customer feedback on food products. Reviews share details about product quality, taste, value, packaging, buying experiences and other parts of how happy customers.

The data gives a foundation for:

- identifying words and themes;
- analysing customer sentiment;
- identifying positive and negative experiences;
- developing an automated sentiment classification model;
- generating business recommendations.

2.3 Data Quality

The data has customer text so it shows marks of real text such as duplicate reviews, punctuation, HTML tags, common stopwords and different writing styles.

Therefore the analysis cleaned the data before running machine learning.

Missing values and duplicate records were. The review text was cleaned before making features.

3. Data Cleaning and Exploratory Data Analysis

3.1 Data Cleaning

Many preprocessing steps were used to ready the review data for analysis.

```
#Check for missing values
missing_values = df.isnull().sum()

# Display the missing values
print(missing_values)
```

Id	0
ProductId	0
UserId	0
ProfileName	26
HelpfulnessNumerator	0
HelpfulnessDenominator	0
Score	0
Time	0
Summary	27
Text	0
dtype:	int64

The main steps included:

1. Checking the structure of the data set.
2. Checking for missing values.
3. Checking for records.
4. Removing missing review text.

5. Creating sentiment labels.
6. Converting text to lowercase.
7. Removing HTML tags.
8. Removing punctuation and special characters.
9. Removing spaces.
10. Removing English stopwords.

After cleaning a new field called Clean_Text was. That field was used for text analysis.

4]:

	Text	Clean_Text
0	I have bought several of the Vitality canned d...	bought several vitality canned dog food produc...
1	Product arrived labeled as Jumbo Salted Peanut...	product arrived labeled jumbo salted peanuts p...
2	This is a confection that has been around a fe...	confection around centuries light pillowy citr...
3	If you are looking for the secret ingredient i...	looking secret ingredient robitussin believe f...
4	Great taffy at a great price. There was a wid...	great taffy great price wide assortment yummy ...

3.2 Sentiment Classification

```

Sample cleaned reviews:
443593    high quality dog food meat fruits grains dog l...
136949    love cake mix mixes well incredible amazon off...
520459    nice strong brew new keurig lived french press...
219515    found pb pb chocolate thrilled love sugary goo...
471273    delightful mint tea one would expect note tea ...
Name: Clean_Text, dtype: str

Sample sentiment labels:
443593    Positive
136949    Positive
520459    Positive
219515    Positive
471273    Positive
Name: Sentiment, dtype: str

Total reviews used for modelling: 50000

```

For sentiment analysis customer ratings were turned into two groups:

- Ratings of 1 and 2 were marked as **Negative**.
- Ratings of 4 and 5 were marked as **Positive**.
- Neutral reviews were left out of the binary model.

The final data set had about **443,777 reviews**. *82,037 Negative reviews*.

That means the data leans heavily toward reviews. This imbalance matters when reading the model results because a model can look very accurate yet still miss the group.

3.3 Review Length Analysis

After cleaning the length of each review was counted. That let the analysis look at how text customers write in their reviews.

Review-length analysis helps because very short reviews give little detail while longer reviews can hold many opinions, product details and explanations of customer experiences.

3.4 Word Frequency Analysis

Word-frequency analysis used Python's Counter. That method was chosen because putting all reviews into one text string would use much memory on the computer.


```

Sample cleaned reviews:
443593    high quality dog food meat fruits grains dog l...
136949    love cake mix mixes well incredible amazon off...
520459    nice strong brew new keurig lived french press...
219515    found pb pb chocolate thrilled love sugary goo...
471273    delightful mint tea one would expect note tea ...
Name: Clean_Text, dtype: str

Sample sentiment labels:
443593    Positive
136949    Positive
520459    Positive
219515    Positive
471273    Positive
Name: Sentiment, dtype: str

Total reviews used for modelling: 50000

```

4.1 Need, for Feature Engineering

Machine-learning algorithms cannot read customer-review sentences directly. The text must first be turned into numbers.

TF-IDF was chosen to do this.

4.2 TF-IDF

Term Frequency-Inverse Document Frequency (TF-IDF) measures how important words are in a document while lowering the weight of words that appear often in the set.

The analysis used:

- Maximum features: **5,000**
- N-gram range: **1–2**
- document frequency: **2**
- Maximum document frequency: **0.95**

Using words and word pairs lets the model capture both individual words and two-word phrases.

For example a single term like *taste* and a phrase like ** taste** can both become features.

4.3 Modelling Sample

Although the full dataset has 568,454 reviews the machine-learning stage used a sample of **50,000 reviews**.

This was needed because TF-IDF creates a matrix and the computer ran into memory limits when handling the whole set.

The sample was chosen with a fixed state of 42 so that the analysis could be repeated.

The 50,000 reviews were split into:

- **40,000 training reviews (80%)**
- **10,000 testing reviews (20%)**

Stratified sampling was used to keep the distribution of sentiment classes.

4.4 TF-IDF Results

The TF-IDF process produced:

- Training matrix: **40,000 × 5,000**
- Testing matrix: **10,000 × 5,000**
- Total features: **5,000**

5. Model Training

5.1 Logistic Regression

Logistic Regression was chosen as the sentiment-classification model.

It is suitable because sentiment classification is a problem and Logistic Regression works with high-dimensional sparse text features like TF-IDF.

The model was configured with:

- $C = 1.0$
- Maximum iterations = 1,000
- state = 42

The model was trained on the TF-IDF representation of the training reviews.

The notebook confirms that the Logistic Regression model was trained successfully.

5.2 Sentiment Predictions

After training the model was used on the 10,000 test reviews.

The model produced a sentiment prediction for each test review.

The first predictions show that the model can classify reviews as either Positive or Negative.

```
# Display confirmation
print("Logistic Regression model trained successfully.")

Logistic Regression model trained successfully.
```

6. Model Evaluation

The Logistic Regression model was evaluated with accuracy, precision, recall and F1-score.

The results from the notebook are shown below.

Metric	Result
--------	--------

Accuracy	91.57%
----------	---------------

Precision	92.07%
-----------	---------------

Recall	98.48%
--------	---------------

F1-score	95.17%
----------	---------------

These are the results generated by the completed Logistic Regression model.

```
# Display the results
print("MODEL PERFORMANCE")
print("-----")
print(f"Accuracy : {accuracy:.4f}")
print(f"Precision: {precision:.4f}")
print(f"Recall   : {recall:.4f}")
print(f"F1-score  : {f1:.4f}")

MODEL PERFORMANCE
-----
Accuracy : 0.9157
Precision: 0.9207
Recall   : 0.9848
F1-score : 0.9517
```

6.1 Accuracy

The model achieved an accuracy of **91.57%**.

This means that 91.57% of the test reviews were classified correctly.

Accuracy gives an measure but it should not be looked at alone because the dataset has many more positive than negative reviews.

6.2 Precision

The model achieved a precision of **92.07%**.

Precision measures how many of the reviews predicted as positive were actually positive.

This shows that the model was generally good, at finding reviews without giving false positives.

6.3 Recall

The model achieved a recall of **98.48%**.

Recall measures how many of the reviews were correctly identified by the model. The high recall shows that the model has been able to find most of the reviews, in the test data.

6.4 F1-score

The model achieved an F1-score of **95.17%**.

The F1-score brings together precision and recall into one measure. Is useful when it is important to consider both kinds of classification error.

The high F1-score indicates that the model gives an overall performance when it classifies the test sample.

7. Topic Modelling

I have observed that sentiment classification only tells us if a review is positive or negative. It does not explain the topics customers talk about.

Therefore topic modelling is needed to find recurring themes inside the review collection.

Latent Dirichlet Allocation (LDA) was chosen for this purpose.

The planned LDA process uses:

- Bag-of-Words representation;
- a review sample to manage memory;
- five topics;

```
Topic modelling data shape: (10000, 3000)
Number of vocabulary terms: 3000
```

-
- a maximum of 10 training iterations;.
- A fixed random state of 42.

The LDA model will find groups of words that often appear together. These groups can then be seen as customer-feedback themes.

The final topic names must be based on the words produced by the Python LDA output not guessed

Potential business themes might be product taste, product quality, price/value, packaging or delivery. These should only be mentioned if the real LDA results support them.

8. Model Improvement

```
IMPROVED MODEL PERFORMANCE
-----
Accuracy : 0.9204
Precision: 0.9299
Recall   : 0.9794
F1-score : 0.9540
```

Model improvement is a part of the analysis because the initial Logistic Regression model can be made better by changing parameters or using modelling approaches.

The initial model used: **C = 1.0**

A second Logistic Regression model can be trained with a regularisation parameter for example: **C = 2.0**

The improved model should be evaluated using the four metrics:

- Accuracy
- Precision
- Recall
- F1-score

The two models should then be compared.

The model-improvement section must report the results after the improved model is run. No improvement should be claimed until those results are available.

9. Key Findings

The analysis provides findings.

First the Amazon Fine Food Reviews dataset offers an relevant source of customer feedback. The dataset contains 568,454 reviews letting the analysis look at a range of customer experiences.

Second the sentiment labels show an imbalance between negative reviews. There are positive reviews than negative reviews. This should be kept in mind when looking at the classification results.

Third text preprocessing is important because customer reviews contain punctuation, HTML elements, stopwords and other unnecessary text features. Cleaning the reviews gave an input for machine learning.

Fourth TF-IDF successfully turned the reviews into features. The final representation had 5,000 features. Included both unigrams and bigrams.

Finally the Logistic Regression model produced classification results on the 10,000-review test set. The model achieved 91.57% accuracy and an F1-score of 95.17%.

10. Business Recommendations

10.1 Automate Customer Feedback Monitoring

The sentiment model can be put into a customer-feedback monitoring process.

New reviews could be automatically classified as positive or negative letting the company track changes in customer sentiment without having to read every review by hand.

10.2 Monitor Feedback

Negative reviews should get attention because they can reveal potential product or service problems.

The company could create a dashboard showing:

- number of reviews;
- percentage of reviews;
- common negative terms;
- changes in negative sentiment over time.

10.3 Combine Sentiment with Topics

Sentiment does not explain why customers are happy or unhappy.

The company should therefore combine sentiment classification with topic modelling.

For example if the LDA model finds a topic about packaging and that topic shows up often in reviews management could look into packaging quality.

10.4 Identify Positive Product Characteristics

Frequently occurring terms, in reviews can help the company see which characteristics customers value.

These insights could support:

- product development;
- marketing;
- product descriptions;
- Customer-retention strategies.

10.5 Investigate Repeated Complaints

Recurring negative themes should be prioritised for investigation. If customers repeatedly mention the product characteristic the company should look into whether the issue's, about product quality, taste, value, packaging, delivery or another problem found by the analysis.

10.6 Continue Monitoring Model Performance

The model should not be seen as accurate.

Customer language changes over time and new products may bring in words that were not in the training data.

The company should therefore. Evaluate the model from time, to time using reviews.

11. Ethical Considerations

Text analytics of customer reviews should consider fairness and bias issues.

The dataset shows customers who chose to submit reviews and may not show all customers.

Customers who are very satisfied or very dissatisfied may be more likely to leave reviews.

The sentiment model may also make mistakes when reviews contain sarcasm, odd language spelling mistakes or complex opinions.

The company should therefore avoid treating automated sentiment predict.

References

- Delen, D., & Demirkan, H. (2020). Big data analytics methods: A 2nd edition. Springer.
- McAuley, J., & Leskovec, J. (2013). Hidden factors and hidden topics: Understanding rating dimensions with review text. Proceedings of the 7th ACM Conference on Recommender Systems, 165–172. <https://dl.acm.org/doi/10.1145/2507157.2507163>
- <https://www.kaggle.com/datasets/snap/amazon-fine-food-reviews?>